

Value Change Record Module (VCRM)

Value Change Record Module (VCRM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VFM® — Value Flow Mechanisms Architecture

Parent Standard: Value Change Standard (VCS)

Module: VCRM — Value Change Record Module

Operational Layer: Value Change Record Layer

Category: Governance & Enforcement

Subcategory: Value Lifecycle Governance

Type: Core Module

Governed Space: Value Change Record

Origin ID: VFM-VCS-VCRM-006.5

Version: 1.0

Status: Canonical · Open Standard

Effective Date: 25 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · OBIDENITY®

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodology Intellectual Property Protection

Canonical Language: UCL — Universal Canonical Language

Canonical Definition

Value Change Record Module (VCRM) is the fifth Core Module of the Value Change Standard (VCS) within VFM® — Value Flow Mechanisms Architecture.

VCRM governs the creation, structuring, preservation, linkage, versioning, and lifecycle continuity of the governed record representing a determined Value Change.

The Module consolidates the governed outputs produced through the preceding VCS modules into a coherent and reconstructable Value Change Record.

The record may preserve:

what changed,

what type of change occurred,

how materially value changed,

what causal basis is associated with the change,

when the change occurred or became determinable,

what evidence supports the determination,

what uncertainty remains,

and

what resulting Value State or subsequent lifecycle process the change may affect.

VCRM does not independently determine whether Value Change occurred, classify the change, calculate its magnitude, or establish its causal basis.

Its task is:

to preserve the governed Value Change as a persistent, traceable, context-bound, versionable, and lifecycle- usable record.

Architectural Position

VCRM is the fifth and final module within VCS.

It follows:

VCCM — Value Change Causation Link Module.

The internal sequence is:

Governed Value State → VCIM — Value Change Identification → VCTM — Value Change Type → VCMM — Value Change Magnitude → VCCM — Value Change Causation Link → VCRM — Value Change Record

VCRM closes the internal governance flow of VCS by transforming the preceding determinations into a persistent governed record.

Internal Module Flow

Identified Value Change → Value Change Type → Value Change Magnitude → Value Change Causation Link → Record Object Creation

→ Change Context Assembly → Temporal Anchoring → Evidence & Determination Linkage → Uncertainty & Limitation Preservation → Record Version Assignment → Lifecycle Relationship Linkage → Governed Value Change Record

Operational Space

VCRM governs the space between:

completed Value Change determination

and

persistent lifecycle representation of that change.

Its operational space includes:

Value Change record identification; record creation; record structure; change-object linkage; value-context linkage; previous-state linkage; resulting-state linkage; change-type preservation; magnitude preservation; causal-link preservation; temporal anchoring; evidence linkage; determination provenance; uncertainty preservation; limitation preservation; record versioning; correction; supersession; lifecycle continuity; record-to-record relationships; change history reconstruction; downstream governance availability.

Value Change ≠ Value Change Record

A change may occur before it is formally recorded.

Therefore:

Value Change ≠ Value Change Record.

VCRM does not create the underlying change.

It creates the governed representation through which the change can remain identifiable and usable across the value lifecycle.

This distinction is fundamental.

A record must represent reality.

It must not substitute for reality.

Value Change Record Object

Each governed Value Change should be representable as a distinct Value Change Record Object.

The record should be capable of identifying, where applicable:

Value Context

Value-Bearing Object

Prior Value State

Value Change

Value Change Type

Value Change Magnitude

Causation Link

Resulting Value State

Temporal Basis

Evidence Basis

Determination Basis

Uncertainty

Limitations

Record Version

Record Status

This creates a persistent governance object rather than an isolated analytical observation.

Record Identity

A Value Change Record should have a stable identity sufficient to distinguish it from other Value Change Records.

Identity may include:

unique record identifier; architecture reference; Parent Standard reference; module reference; governed value reference; value-bearing object reference; temporal reference; version; record status.

Record identity enables Value Change to remain referencable across systems and over time.

Value Context Linkage

Every Value Change Record must remain connected to the governed Value Context established under:

Value Context Standard (VCXS).

A statement such as:

Value decreased by 20%.

is insufficient without knowing:

which value,

of what,

for whom,

for what purpose,

and

against what reference context.

VCRM therefore preserves the contextual identity of the changed value.

Previous Value State Linkage

Where applicable, the record should identify the Value State existing before the change.

For example:

Prior State: Productive Value Asset

→ Change: Physical deterioration

→ Resulting State: Reduced productive value.

This creates a traceable relationship between:

State Before Change

and

Change Event.

Resulting Value State Linkage

Where the resulting Value State has been determined, VCRM should preserve that relationship.

The lifecycle relationship becomes:

Prior Value State → Governed Value Change → Resulting Value State.

VCRM does not independently determine the resulting state.

That determination remains governed through the appropriate Value State governance.

The record preserves the relationship.

Change Type Preservation

VCRM receives the governed Value Change Type from VCTM.

The record may preserve classifications such as:

increase; decrease; creation;

reduction; degradation; enhancement; impairment; recovery; appreciation; depreciation; conversion-related change; partial loss; complete loss; other governed change types.

The record must preserve the classification actually determined rather than silently reclassifying it downstream.

Magnitude Preservation

VCRM receives the governed Value Change Magnitude from VCMM.

Magnitude may be:

quantitative; qualitative; proportional; monetary; operational; physical; capability-based; performance-based; multi-dimensional; conditionally determined.

Where magnitude contains uncertainty, range, confidence, methodology, baseline, or reference conditions, those properties should remain connected to the record.

A stored number without its basis may lose its governed meaning.

Causation Link Preservation

VCRM receives the governed causal relationship from VCCM.

The record may preserve:

causal factor; causal basis; causal chain; contributing causes; alternative causes; causal contribution; causal-link strength; uncertainty; contested causation; unresolved causation.

VCRM must not transform:

Possible Cause

into

Established Cause

merely because the relationship has entered a formal record.

Temporal Anchoring

Value Change requires a temporal basis.

Where possible, VCRM should distinguish between:

Occurrence Time

when the underlying change occurred;

Detection Time

when the change became observable;

Determination Time

when the change was formally determined;

Recording Time

when the governed record was created.

These timestamps may differ materially.

For example:

Value deterioration occurred gradually over six months

but
was detected during a quarterly review
and
formally determined two weeks later.
Collapsing these moments into one timestamp can distort lifecycle analysis.

Point Change and Period Change

VCRM supports both:

Point-in-Time Change

and

Change Across a Period.

A payment may create a relatively discrete Value Change.

Physical deterioration, capability degradation, market decline, or operational improvement may occur progressively.

The record should therefore preserve whether the change represents:

event-based change; interval-based change; cumulative change; continuous change; progressive change; episodic change.

Evidence Linkage

A Value Change Record should remain linked to the evidence supporting its governed determinations where such evidence exists.

Evidence may include:

measurements; operational data; transaction records; market observations; technical inspection; sensor information; accounting information; contractual information; simulation outputs; valuation analysis; expert determination; external authority determination; another governed evidential basis.

VCRM does not validate the evidence itself.

It preserves the relationship between the record and the evidential basis upon which the relevant determination relies.

Determination Provenance

A governed record should preserve sufficient provenance to reconstruct how its material determinations entered the record.

This may include:

determination source; methodology; responsible process; relevant system; external determination; date; version; applicable conditions; referenced evidence; superseded determination.

The objective is not to turn VCRM into an audit architecture.

The objective is to prevent Value Change records from becoming detached from the basis that gives them meaning.

Uncertainty Preservation

Uncertainty must travel with the Value Change Record.

Examples include:

Magnitude: CHF 8–11M reduction

rather than silently storing:

Magnitude: CHF 9.5M.

Or:

Causal Link: Supported

rather than:

Cause: Established.

VCRM preserves uncertainty generated by upstream governance.

It does not erase uncertainty to make records appear simpler.

Limitation Preservation

Material limitations affecting interpretation of the Value Change should remain attached to the record.

Limitations may relate to:

incomplete evidence; partial observation; uncertain baseline; unresolved causation; model dependence; limited measurement; missing period; conditional magnitude; contested determination; scope restriction; temporal uncertainty.

A record without its material limitations may become misleading even if every stored number is technically correct.

Record Status

A Value Change Record may have a governed status.

A minimum model may include:

Draft

Active

Under Review

Contested

Corrected

Superseded

Archived

Status must remain distinguishable from the underlying Value Change itself.

Record Status ≠ Value State.

Record Versioning

Value Change records may evolve when new information becomes available.

VCRM should therefore support versioning.

For example:

Record V1

Initial magnitude:

–CHF 8M to –CHF 12M

Causal basis:

Indeterminate

Later evidence produces:

Record V2

Magnitude:

–CHF 9.4M

Causal basis:

Supported multi-causal relationship.

The new version should not silently erase the previous record.

The lifecycle should preserve:

what was known,

when it was known,

and

what changed in the determination.

Correction ≠ Historical Erasure

Where a record contains an error, VCRM should support correction while preserving appropriate historical continuity.

Therefore:

Correction ≠ Deletion of Prior Governance History.

A corrected record may identify:

previous value; corrected value; correction basis; correction time; affected determination; resulting version.

This enables reconstruction of decisions made using earlier information.

Supersession

A Value Change Record may be superseded where a later governed determination replaces it.

Supersession should identify:

Superseded Record

→ Supersession Basis

→ New Record

→ Effective Relationship.

Supersession does not imply that the earlier record was necessarily erroneous.

It may simply reflect better evidence or a later determination.

Record Immutability and Controlled Evolution

VCRM does not require one universal technical implementation.

Organizations may use:

append-only records; version-controlled databases; event stores; relational systems; distributed ledgers; enterprise systems; governed data platforms; hybrid infrastructures.

The methodological requirement is that material history remain reconstructable.

Therefore:

Technical Mutability may vary.

Governance History must remain controlled.

Record Granularity

VCRM should preserve sufficient granularity to represent materially distinct Value Changes separately.

For example, an enterprise may experience:

+CHF 20M technological capability gain

and

–CHF 12M market-access deterioration

during the same period.

Recording only:

Net Value Change: +CHF 8M

may destroy materially relevant lifecycle information.

VCRM therefore supports separate change records that may later be aggregated where appropriate.

Aggregation ≠ Replacement

Multiple Value Change Records may be aggregated for:

portfolio analysis; enterprise reporting; investment assessment; asset management; banking; insurance; infrastructure management; mission management; strategic analysis.

However:

Aggregated Value Change ≠ Replacement of Constituent Change Records.

The underlying records should remain traceable where materially relevant.

Record Relationships

Value Change Records may have relationships such as:

sequential; concurrent; cumulative; causal; compensating; offsetting; dependent; cascading; reversing; recovery-related; transformation-related.

These relationships enable VFM® to represent complex value histories rather than isolated snapshots.

Value Change Chain

VCRM enables reconstruction of a Value Change Chain.

For example:

Value State A → Change Record 001: +15% → Value State B → Change Record 002: -8% → Value State C → Change Record 003: -100% → ZVA™ — Zero Value Asset™ state

The chain allows the current Value State to be understood through the sequence of governed changes that produced it.

Lifecycle Continuity

VCRM creates continuity between Value Change and subsequent VFM® governance.

A Value Change Record may become an input to:

Value State reassessment; value transition governance; value flow analysis; value loss analysis; value recovery analysis; monetization mechanisms; de-monetization mechanisms; re-monetization mechanisms; portfolio aggregation; future lifecycle standards.

This is why VCRM is not merely documentation.

It is a lifecycle interface.

Machine-Readable Record Principle

VCRM should support machine-readable representation where implementation permits.

A machine-readable Value Change Record may enable systems to identify:

Object X

Value Context Y

Previous State A

Change Type: Decrease

Magnitude: -17%

Cause Link: Supported

Resulting State B

Effective Period: T1-T2

Uncertainty: Defined

Record Version: 3

This enables automated reasoning across value histories without collapsing governance context.

Cross-System Portability

A Value Change Record should remain interpretable when transferred between systems where legitimate.

Portability requires preservation of material semantics.

A transferred record should not lose:

value context; object reference; state relationship; type; magnitude basis; causal-link state; temporal basis; uncertainty; limitations; version.

The record may change technical format.

Its governed meaning should not silently change.

VCRM Boundary with VCIM

VCIM determines:

whether Value Change occurred.

VCRM records the governed result.

Therefore:

Change Identification $\neq$ Change Recording.

VCRM Boundary with VCTM

VCTM determines:

what type of Value Change occurred.

VCRM preserves that determination.

Therefore:

Classification $\neq$ Record.

VCRM Boundary with VCMM

VCMM determines:

the magnitude of Value Change.

VCRM preserves the magnitude and its basis.

Therefore:

Magnitude Determination $\neq$ Magnitude Record.

VCRM Boundary with VCCM

VCCM governs:

the causal relationship associated with Value Change.

VCRM preserves that causal relationship without independently establishing it.

Therefore:

Causation Link $\neq$ Change Record.

VCRM Boundary with VSS

VSS governs:

the current governed Value State.

VCRM governs:

the persistent record of a Value Change.

Therefore:

Value State $\neq$ Value Change Record.

The two may be linked:

Prior Value State $\rightarrow$ Value Change Record $\rightarrow$ Resulting Value State.

VCRM Boundary with OBIDENTITY®

VCRM may reference the identity of the value-bearing object.

It does not establish universal object identity, ownership, origin, authority, or permission.

Where those relationships require governance, OBIDENTITY® remains the relevant architecture.

VCRM Boundary with INTEGROS®

VCRM requires records to preserve the information necessary for their governed meaning.

It does not replace general integrity governance.

INTEGROS® may govern integrity properties affecting the record, evidence, state, or supporting systems.

VCRM Boundary with ADIT®

VCRM creates and maintains the governed Value Change Record.

ADIT® may govern auditability, evidential preservation, examination, verification, or reconstruction.

Therefore:

Record Governance $\neq$ Audit Governance.

Minimum Implementation Framework

1. Record Object Creation

Create a uniquely identifiable Value Change Record.

2. Value Context Linkage

Link the record to the governed value and relevant value-bearing object.

3. Prior State Linkage

Identify the preceding governed Value State where applicable.

4. Change Determination Assembly

Preserve the outputs of VCIM, VCTM, VCMM, and VCCM.

5. Temporal Anchoring

Record relevant occurrence, detection, determination, and recording times or periods.

6. Evidence & Provenance Linkage

Preserve links to supporting evidence and determination provenance.

7. Uncertainty & Limitation Preservation

Attach material uncertainty, conditions, and limitations.

8. Resulting State Linkage

Link the record to the resulting governed Value State where determined.

9. Version & Status Control

Assign record version, status, correction, and supersession relationships.

10. Lifecycle Availability

Make the governed record available to authorized downstream VFM® processes without loss of material meaning.

Governance Outputs

VCRM may produce:

Value Change Record Value Change Record ID Value Change Context Link Prior Value State Link Resulting Value State Link Value Change Type Record Value Change Magnitude Record Value Change Causation Link Record Value Change Temporal Record Value Change Evidence Link Value Change Determination Provenance Record Value Change Uncertainty Record Value Change Limitation Record Value Change Version Record Value Change Supersession Record Value Change Chain Governed Value Change Record Object

Use Case 1 — Banking: Reconstructable

Collateral Value History

A bank monitors an industrial asset used as collateral.

During three years the asset experiences several materially different Value Changes:

Year 1: technological upgrade increases productive capability.

Year 2: major customer loss reduces operating value.

Year 3: physical deterioration reduces productive capacity.

A conventional system may preserve only periodic valuations:

CHF 30M → CHF 34M → CHF 27M → CHF 21M.

VCRM preserves the governed lifecycle behind those numbers:

Value State A → Upgrade Change Record → Value State B → Customer-Loss Change Record → Value State C → Deterioration Change Record → Current Value State D.

Each record retains its:

change type, magnitude, causal basis, evidence, temporal context, uncertainty, and version.

The bank can therefore distinguish between value deterioration caused by:

market conditions,

operational deterioration,

and

physical asset degradation.

This supports stronger:

credit monitoring, collateral governance, impairment assessment, restructuring analysis, recovery planning, and portfolio risk analysis.

The institution no longer sees only:
what the asset is worth today.
It can reconstruct:
how the asset arrived at its current governed Value State.

Use Case 2 — Space Mission: Preserving Mission

Value Changes Across Operations

A long-duration space mission has an initial governed mission capability and value basis.

During operations:

solar-array efficiency falls;
propellant consumption exceeds plan;
a software optimization reduces computational demand;
a scientific instrument develops a partial fault;
mission planning recovers part of the lost operational capability.

These changes do not move mission value in one direction.

Some destroy value.

Some preserve value.

Some recover value.

Some affect only one mission objective.

VCRM creates separate governed records:

Solar Degradation Record

Propellant Consumption Record

Computational Optimization Record

Instrument Degradation Record

Mission Recovery Record.

Each record remains connected to:

the relevant mission Value Context,
the prior Value State,
the magnitude of change,
the causal basis,
the applicable mission period,
the evidence supporting the determination,
and
the resulting Value State.

Mission systems can therefore reconstruct:

Initial Mission Value → Value Loss → Further Value Loss → Efficiency Gain → Capability Degradation → Value Recovery → Current Mission Value State.

This allows future mission decisions to be based not merely on the current state but on the governed history of how mission value evolved.

Enterprise Application

VCRM enables organizations to build a persistent Value Change history rather than maintaining disconnected valuations, measurements, observations, or reports.

It can support:

banking; insurance; investment; asset management; industrial production; infrastructure; energy; AI systems; computational infrastructure; blockchain and digital assets; intellectual property; supply chains; mission systems; public assets; organizational capability; technology portfolios.

Its strategic importance increases where organizations need to understand not merely:

What is the value now?

but:

How did the value get here?

A governed Value Change history can support future analysis of:

value creation,

value destruction,

value preservation,

value recovery,

causal patterns,

transition mechanisms,

and

value lifecycle performance.

Foundational Principles

Value Change is not the same as a Value Change Record.

A record represents change; it does not create the underlying change.

A Value Change Record must remain connected to its Value Context.

Current Value State should be reconstructable from materially relevant prior Value Changes where the evidence permits.

Change type, magnitude, causation, uncertainty, and limitations must not lose their governed meaning when recorded.

Occurrence time, detection time, determination time, and recording time may be different.

Correction must not require historical erasure.

Supersession must remain traceable.

Aggregation must not destroy materially relevant constituent changes.

A record version must preserve what was known at that point in the lifecycle.

Value Change history is a governed lifecycle asset, not merely administrative documentation.

Canonical Closing Statement

Value Change Record Module establishes the persistent record layer of the Value Change Standard by transforming identified, classified, magnitude-determined, and causally linked Value Change into a structured, traceable, context-bound, temporally anchored, evidence-linked, uncertainty-preserving, version-controlled, and lifecycle-usable governed record. Through governance of record identity, Value Context linkage, prior and resulting Value States, change type, magnitude, causation links, temporal basis, evidence, provenance, uncertainty, limitations, status, correction, supersession, record relationships, aggregation, portability, and lifecycle continuity, VCRM enables Value Change to remain reconstructable after the immediate determination has ended. By separating the reality of Value Change from its recorded representation, and by preserving the outputs of the preceding VCS modules without silently redefining them, VCRM creates the durable governance bridge between individual Value Changes and the wider VFM® value lifecycle. A governed value system should not merely know the current Value State; where materially possible, it should be able to reconstruct the governed sequence of changes through which that state emerged.

Governed Links

[→ Value Change Standard \(VCS\)](#)

[→ VFM® Architecture Map](#)

[→ Previous: Value Change Causation Link Module \(VCCM\)](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>